Fiscal Year:	FY 2021	**Task Last Updated:**	FY 06/22/2021
PI Name:	Froio-Blumsack, Danielle M.S.		
Project Title:	Stabilized Foods for Use in Extended Spaceflight: Preservation of Shelf-Life, Nutrient Content and Acceptability		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	United States Department of the Army		
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City:	Natick	**State:**	MA
Zip Code:	01760	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	04/21/2014	**End Date:**	10/31/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2022 per C. Ribeiro/HHC element/JSC HRP (Ed., 11/18/21		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ13HA911 IAA		
Performance Goal No.:			
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Task Description:	ED. NOTE (June 2021): Principal Investigator (PI) Danielle Froio-Blumsack continues work initiated with PI Dr. Ann Barrett, whose work covered the first 3 years of this contract with U.S. Army. The objective of this effort is to develop shelf stable, highly acceptable food with increased nutrient (vitamins) stability for extended space missions utilizing innovative processing and packaging technologies. Selected packaging technologies will be investigated with research focused on the interaction of packaging material with various innovative sterilization processes such as microwave heating, irradiation, and high pressure treatment. The availability of highly nutritious and health-promoting food is a factor that is a significant prerequisite for prolonged space travel. The design of feeding and nutritional strategies for multi-year, non-resupplied flights is an undertaking requiring substantial research and development; it is also an endeavor and that could be founded upon our existing, considerable knowledge and experience base at the U.S. Army Combat Capabilities Development Command Soldier Center.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study will yield strategies for the development of extremely stable, nutrient-dense foods and the development of packaging materials compatible with new quality-preserving sterilization techniques. While this work is specifically important to the health of astronauts, its significance also extends to the research that is critical to the mission of the Natick Soldier Research Development and Engineering Center: to support and promote the nutritional health of the Warfighter on extended missions with little or no means of resupply.
Task Progress:	Effort 2 - Packaging Thrust Effort. 48 Month Report. Introduction: The most common commercial sterilization process for high water activity, low-acid foods is retorting. However, retort exposes product to prolonged high temperatures and can severely reduce food quality and package integrity. Innovative packaging materials and preservation technologies, including microwave-assisted thermal sterilization (MATS) and irradiation, are being explored in an effort to enhance quality, vitamin stability, and shelf-life. This investigation is supported by NASA in an effort to develop highly stable foods for long-term space exploration. The objective of this effort was to investigate packaging materials for compatibility with novel food sterilization methods and assess food quality and vitamin stability as a function of processing method and package type. Experimental: A non-foil polymer-based pouch utilizing a high barrier aluminum oxide coating (Pouch A) and a control aluminum foil based pouch (Pouch B) were evaluated for their compatibility with advanced processing methods. Pouches contained a Creamy Cajun Chicken (CCC) formulation fortified with two times the spaceflight requirement of vitamins A, B1, B9, C, and E; they were subjected to retort, irradiation, or microwave assisted thermal sterilization (MATS) processing. Samples were stored for 4 years at 70oF, and pulled and evaluated at yearly increments. Microbiological testing was conducted at each pull to confirm sustained commercial sterility. Sensory analysis and vitamin analysis were conducted. Pouch integrity was described by Young's modulus, seal strength, and burst strength. Barrier performance was measured in terms of oxygen transmission rate (OTR) and water vapor transmission rate (WVTR). Conclusions Retention of vitamins E, B9, and A was relatively unchanged over the four-year storage study for MATS and retorted products. Conversely, all processing methods resulted in product that showed a significant decrease in vitamins B1 and C after four years; however, MATS showed higher concentration of these two vitamins in comparison to other processing methods. A decrease in Young's modulus, seal strength, and burst strength was observed for both pouches, but did not impact overall mechanical integrity of the pouch under normal handling. Evaluating pouches for rough handling at the conclusion of the extended storage study could offer insight into how loss of mechanical properties impacts pouch performance during worst case scenarios. OTR and WVTR fluctuated over the course of the study, but did not show an increasing trend over storage time, nor did it appear to impact sensory attributes of the food product (during 3-year storage).
Bibliography Type:	Description: (Last Updated: 06/22/2021)
Abstracts for Journals and Proceedings	Froio DF, Barrett AH, Richardson MJ. "The effect of novel processing methods on vitamin stability, food quality and package integrity." Poster Presentation at Institute of Food Technologists (IFT) Annual Conference 2018, Chicago, IL, July 15-18, 2018. Abstracts. Institute of Food Technologists (IFT) Annual Conference 2018, Chicago, IL, July 15-18, 2018. , Jul-2018
Abstracts for Journals and Proceedings	Froio DF, Barrett AH, Richardson MJ. "The effect of novel processing methods on vitamin stability, food quality and package integrity." Poster Presentation at Institute of Food Technologists (IFT) Annual Conference 2020, Virtual, July 13, 2020. Institute of Food Technologists (IFT) Annual Conference 2020, Virtual, July 13, 2020. , Jul-2020